Microsoft Azure DevOps: The Rapid Software Development Platform

Azure DevOps is an amazing offering from Microsoft that lets users manage projects that are platform-, language-, and operating system-agnostic. With Azure DevOps pipelines, it is very easy to manage project task items, CI/CD pipelines, and even publishing of artifacts. This article begins with the basic concepts and jargon associated with Azure DevOps, moves on to build a simple pipeline, and enhances it by leveraging variables.

Back in the 1990s, industries often followed the waterfall model for software development, where each department had a separate responsibility like requirements, development, test and operations. In such a model, once a change is made, it becomes difficult to backtrack and modify something. So the model is not good for projects that have frequent requirement changes.

The modern approach is for industries to follow the agile method of work, which removes the strict separation between the requirements, development, and test teams. The project teams are generally proficient in all disciplines. Consequently, the development of software products is speeded up as changes in requirements or errors can be detected and remedied as they occur.

DevOps is the practice that brings the development and operations teams together. It's all about creating a repeatable and reliable process for delivering software to customers fast. The DevOps toolchain is used to achieve continuous integration and continuous deployment. DevOps also helps teams to identify errors quickly, as the development team can easily check the latest release builds.

Why is Azure DevOps needed?

We need a toolset to implement DevOps, which provides insights about all the steps involved in the building and running of software. Microsoft's Azure DevOps software suite has all the tools necessary for doing so.

Users can also opt for a set of independent tools. For instance, for the agile scrum boards, Jira or Trello may be used. Build tools like Maven, TeamCity, and Bamboo can also be deployed. There are also several tools for release management like XebiaLabs, XL Deploy, Jenkins, Octopus Deploy, etc.

These solutions mean that the tools have to be hosted on-premise and need to be maintained and updated by users themselves. Also, using lots of dispersed tools for DevOps makes it challenging to keep track of all the events happening during the project development.